

Scalable and Robust Named Entity Recognition in Audio Data: A Hybrid Pipeline Approach

Lucas Lamisse^{1*}

¹Master Intelligent Systems and Applications (ISA), Université de Tours, Tours, France.

Corresponding author(s). E-mail(s): lucas.lamisse@gmail.com;

Abstract

The exponential growth of audiovisual content necessitates robust automated Information Extraction systems. While Named Entity Recognition (NER) models achieve near-human performance on well-formatted written text, their accuracy degrades significantly when applied to Automatic Speech Recognition (ASR) transcripts due to the absence of punctuation and structural casing. Consequently, standard models often erroneously merge consecutive entities into semantic chimeras. Furthermore, relying entirely on Large Language Models (LLMs) to process massive audio datasets is computationally prohibitive. In this paper, we propose a highly scalable, hybrid NER pipeline tailored for television broadcast audio. Our approach integrates a heavy baseline—utilizing a customized strict BIO-tag aggregation strategy on CamemBERT alongside a grammatical filter—with the targeted deployment of an LLM (Llama 3) acting as a semantic judge. This dual architecture drastically reduces API calls and computational overhead while effectively disambiguating and splitting complex overlapping entities. We demonstrate the efficiency of our checkpointing batch-processor on a large real-world dataset, establishing a robust methodology for evaluating audio-derived NLP tasks.

Keywords: Named Entity Recognition, Audio Transcription, Large Language Models, Natural Language Processing, Information Extraction, Hybrid Pipeline

1 Introduction

1.1 Context and Motivation

The continuous production of audiovisual content, ranging from television broadcasts to documentaries and investigative journalism, represents a massive and largely unstructured repository of information. To index, archive, and analyze this data at scale, automated Information Extraction (IE) is essential. Named Entity Recognition (NER), the process of identifying and classifying entities such as persons, locations, and organizations, serves as the foundational pillar for downstream tasks like knowledge graph construction and trend analysis.

Traditionally, NER systems have been trained and evaluated on highly structured, grammatically correct written text (e.g., news articles, encyclopedias) where they achieve remarkable precision [1, 2]. However, applying these systems directly to spoken language—transcribed via Automatic Speech Recognition (ASR) models—introduces a severe domain shift. Spoken language is inherently noisy, characterized by hesitations, colloquialisms, and, most critically, a total absence of explicit punctuation.

1.2 Problem Statement

The transition from audio to text systematically breaks the fundamental heuristic rules relied upon by classical NLP models. When an ASR system transcribes an audio stream, it rarely outputs reliable commas or periods. For standard transformer-based NER models like CamemBERT [2], this lack of boundary markers leads to a critical failure known as "entity fusion" or the creation of "chimeras".

For instance, an audio segment mentioning two distinct entities, "*Roméo*" (Person) and "*Buenos Aires*" (Location), spoken without a pause, will be transcribed continuously. Greedy aggregation algorithms typically fail to distinguish the transition, resulting in a fused, erroneous entity (e.g., classifying "*Roméo Buenos Aires*" as a single Location).

While modern Large Language Models (LLMs) possess the world knowledge necessary to recognize and separate such chimeras, deploying them to process raw transcripts for hundreds of hours of video is computationally and financially prohibitive. Conversely, pure algorithmic approaches lack the semantic awareness to correct ASR-induced hallucinations. Thus, a critical bottleneck exists: how to achieve LLM-level semantic accuracy on massive audio datasets without incurring LLM-level computational costs.

1.3 Contributions

To bridge the gap between scalability and semantic accuracy, we propose a novel hybrid pipeline designed to extract robust named entities from noisy broadcast data. Our methodology minimizes computational overhead by using LLMs strictly as a refinement layer rather than a primary extractor.

The main contributions of this paper are as follows:

- **Strict BIO-Aggregation Algorithm:** We introduce a custom aggregation strategy for token classification models that explicitly prevents unwarranted entity fusion at the token level, effectively isolating candidate components.
- **Context-Aware Grammatical Filtering:** We implement a deterministic screening layer that leverages dependency parsing to dynamically eliminate verbs, pronouns, and adverbs masquerading as entities, drastically reducing false positives early in the pipeline.
- **Targeted LLM Disambiguation:** We deploy a generative LLM (Llama 3) strictly as a semantic judge on pre-filtered candidates. This approach successfully splits ASR-induced chimeras while maintaining a minimal computational footprint.
- **Scalable Checkpointing Architecture:** We provide a resilient, fault-tolerant batch-processing framework capable of orchestrating multi-model inference (Source Separation, ASR, NER, LLM) over extensive audiovisual databases.

2 Related Work

2.1 Evolution of Named Entity Recognition

Named Entity Recognition (NER) has undergone a paradigm shift over the past few years, transitioning from traditional statistical methods to deep learning approaches. Transformer-based architectures have established new state-of-the-art baselines across multiple languages by leveraging deep contextualized representations [1, 3–5]. For the French language specifically, models such as CamemBERT and FlauBERT have demonstrated exceptional capabilities in capturing complex semantic relationships within text [2, 6, 7]. However, the vast majority of these models are trained and evaluated on highly normalized, grammatically correct written corpora such as Wikipedia or formal news articles [8–10]. Research consistently shows that when these models are applied to out-of-domain data—particularly conversational or spoken language—their performance degrades significantly [11, 12], highlighting a critical domain gap that necessitates specialized preprocessing or advanced few-shot domain adaptation techniques [13, 14].

2.2 Spoken Language Understanding and ASR Errors

Extracting information from audio sources inherently relies on Automatic Speech Recognition (ASR) systems. The integration of ASR and NER creates a cascading error pipeline where transcription anomalies directly corrupt the downstream entity extraction [15–17]. Modern ASR models, including self-supervised architectures like wav2vec 2.0 and large-scale weakly supervised models like Whisper, have drastically reduced Word Error Rates (WER) [18–21]. Despite these advances, broadcast audio remains challenging due to phonetic confusion, hesitations, and most notably, the complete absence of punctuation and structural casing in raw outputs. To mitigate these issues, recent studies have explored joint ASR-NER training methods [22, 23] or the implementation of robust punctuation restoration modules [24, 25]. Nevertheless, joint models are computationally expensive to fine-tune and often language-specific, while standalone punctuation modules frequently struggle with the unpredictable

disfluencies common in television broadcasts [26]. Consequently, the comparative effectiveness of cascaded extraction pipelines versus purely end-to-end Spoken Language Understanding (SLU) models remains a highly active area of research [27, 28].

2.3 LLMs and Hybrid Extraction Pipelines

The recent advent of Large Language Models (LLMs) has introduced unprecedented zero-shot capabilities for Information Extraction [29, 30]. Models such as GPT-4 and Llama 3 possess the extensive world knowledge required to resolve complex semantic ambiguities and correct ASR-induced hallucinations [31, 32]. Despite these advantages, deploying LLMs to directly process massive, token-heavy audio transcripts is computationally prohibitive and financially restrictive. To balance computational efficiency with semantic accuracy, the literature is increasingly moving towards hybrid architectures. In these systems, smaller, deterministic models handle the bulk of the initial extraction, while LLMs are strategically reserved for complex downstream refinement, logical reasoning, and error mitigation [33]. Our proposed methodology extends this hybrid paradigm, specifically optimizing the baseline filtering process to dramatically reduce the LLM’s workload during scalable audio analysis.

3 Proposed Method

To address the challenges of ASR-induced entity fusion and the computational constraints of processing large audio datasets, we propose a highly scalable, hybrid NLP pipeline. Our architecture operates in four sequential stages: Audio Preprocessing, Heavy Baseline Extraction, Semantic Refinement, and Automated Evaluation.

3.1 Audio Preprocessing and Transcription

The first module of our pipeline transforms raw broadcast video files (MP4) into high-quality textual transcripts. Given the noisy nature of television audio, which often contains background music, sound effects, and overlapping speech, we first apply Source Separation using the Demucs architecture. By specifying a two-stems extraction (*vocals* vs. *accompaniment*), we isolate a clean vocal track, significantly reducing the acoustic noise floor.

The isolated vocal track is then transcribed using the Faster-Whisper framework (Large-v2 model). To optimize transcript boundaries and eliminate non-speech hallucinations, we enable Voice Activity Detection (VAD) filtering with a minimum silence duration of 300 milliseconds. The transcription is forced in French, and we apply a strict compression ratio threshold of 2.6 to prevent the model from entering repetitive, degenerative loops—a known vulnerability of Whisper architectures.

3.2 Heavy Baseline Extraction: Segment-Based Processing and Strict Aggregation

A critical limitation of Transformer-based architectures like CamemBERT is their strict maximum sequence length constraint (typically 512 tokens). Passing an entire

hour-long transcript would result in catastrophic memory truncation, silently dropping the vast majority of the text. To bypass this, we leverage the native timestamped audio segments generated by Faster-Whisper. By processing the transcript chunk-by-chunk, we ensure comprehensive coverage of the broadcast without saturating the model’s memory limits.

The core innovation of our initial extraction phase lies in preventing the formation of semantic chimeras (e.g., *"Roméo Buenos Aires"*). We utilize CamemBERT [2] initialized for token classification. Crucially, we disable the default heuristic aggregation strategies provided by standard NLP libraries by explicitly setting `aggregation_strategy="none"` and `ignore_labels=[]`.

By retaining the raw output tokens, including the `O` (Outside) tags which often represent implicit pauses or silent commas, we implement a custom, strict BIO (Beginning, Inside, Outside) aggregation algorithm. Our deterministic rule dictates that any token tagged with a `B-` prefix strictly initiates a new independent entity object, regardless of spacing. This mathematical separation decapitates multi-category chimeras early in the pipeline.

To further reduce false positives before they reach the LLM, we subject all candidate entities to a grammatical screening layer powered by spaCy. Rather than relying on a static blacklist, we perform contextual part-of-speech (POS) tagging. Candidate words identified grammatically as verbs in imperative forms (e.g., *"Souriez"*), pronouns, or interjections within their respective sentence structure are immediately discarded. Furthermore, universal length constraints are applied (e.g., maximum of 3 words for `PER` and `LOC`, 5 words for `ORG`) to filter out systemic ASR hallucinations.

3.3 Semantic Refinement via Large Language Models

The pre-filtered candidates are structurally valid but may still contain semantic ambiguities that algorithmic rules cannot resolve (e.g., distinguishing *"Direct"* as an adverb versus *"Direct Assurance"* as a company). To resolve this, we deploy Llama 3 [14]—accessed via a high-speed inference API—acting as the ultimate semantic judge.

Instead of passing the entire raw transcript to the LLM, we inject a highly structured prompt containing the original sentence context alongside the drastically reduced list of baseline candidates. The LLM is instructed under a strict ontology constraint to:

1. Evaluate the semantic validity of each candidate based on context.
2. Re-classify the entity if the baseline model assigned an incorrect label.
3. Explicitly split any remaining chimeras into separate JSON objects if ASR completely fused them without `B-` tags.
4. Drop false positives entirely.

To guarantee structural integrity, the LLM is constrained to output its response strictly in JSON format.

3.4 Scalable Batch-Processing and Checkpointing

Processing hundreds of hours of video requires fault-tolerant engineering. Our pipeline is orchestrated by a centralized "Batch Processor" that loads the heavy models (Whisper, CamemBERT, spaCy) into the GPU VRAM only once at initialization.

To mitigate API rate limits (e.g., HTTP 429 Limit Exceeded) encountered during LLM refinement, we implemented an exponential backoff and retry mechanism, pausing the pipeline dynamically when server thresholds are reached. Furthermore, a real-time checkpointing system ensures that every processed audio file is immediately logged in a master JSON registry. This allows the pipeline to safely resume from the exact point of interruption in the event of hardware sleep states or unexpected crashes, making the methodology highly scalable for enterprise-level datasets.

4 Experiments

4.1 Dataset and Ground Truth

To evaluate our proposed pipeline in a real-world scenario, we utilized a custom dataset consisting of diverse French television broadcast audio tracks. These files encompass various challenging acoustic environments, including investigative reports, news broadcasts, and entertainment shows, characterized by overlapping speech and background noise.

Given the impossibility of manually annotating hundreds of hours of video, we established a *Golden Dataset* for a representative subset of the corpus. This ground truth consists of manually verified JSON files (e.g., `Nes_list_*.json`) containing all named entities present in the corresponding broadcasts. The unannotated remainder of the dataset ("in the wild" data) was used to test the pipeline's robustness, scalability, and API rate-limit management during continuous batch processing.

4.2 Evaluation Metrics: Soft-Boundary and Fuzzy Matching

Because ASR transcripts frequently alter the exact spelling of proper nouns (e.g., transcribing "Marie Bonnisseau" as "Marie Bonelisso"), standard strict string-matching evaluation severely and unfairly penalizes the Information Extraction pipeline for acoustic Word Error Rate (WER) anomalies.

To accurately assess the NER model's true semantic retrieval capabilities, we propose a "Fuzzy Matching" evaluation methodology. First, both the predicted entities (P) and the Ground Truth (G) undergo a strict normalization pipeline: removal of accents, lowercasing, and punctuation stripping.

Second, we employ the Levenshtein distance algorithm to calculate a similarity ratio between the predicted and actual entities. We established a strict mathematical threshold of 85% similarity. If an extracted entity matches a Ground Truth entity with a score ≥ 0.85 , and shares the exact same semantic label (e.g., both are PER), it is counted as a valid extraction. The metrics are formulated as follows:

- **True Positives (TP):** Entities matching a ground truth record with $\geq 85\%$ similarity and identical classification tags.

- **False Positives (FP):** Extracted entities that fail to reach the similarity threshold with any ground truth entity (hallucinations or grammatical errors).
- **False Negatives (FN):** Ground truth entities that remain completely unmatched by the pipeline’s output.

We subsequently calculate the standard Information Extraction metrics: Precision ($\frac{TP}{TP+FP}$), Recall ($\frac{TP}{TP+FN}$), and the macro F1-Score.

4.3 Experimental Setup

The initial processing stages—comprising Demucs source separation, Faster-Whisper (Large-v2) transcription, and CamemBERT token classification—were executed locally on a dedicated GPU to eliminate data transfer latency. The semantic refinement stage was handled by the Llama 3.1 8B Instruct model, accessed via the Groq API to leverage its high-throughput inference capabilities. To ensure uninterrupted execution across the 843 audio files, the pipeline was equipped with an automatic checkpointing system and an exponential backoff algorithm to smoothly handle HTTP 429 Rate Limit responses from the API.

4.4 Results and Discussion

The evaluation on the Golden Dataset demonstrates the critical necessity of both the LLM refinement layer and the fuzzy evaluation paradigm. As shown in Table 1, while the baseline models extract a large volume of candidates, their Precision is significantly hampered by grammatical false positives and conversational noise inherent to broadcast audio.

Table 1 Performance Comparison on the Golden Dataset

Model Configuration	Precision	Recall	F1-Score
Baseline (CamemBERT only)	0.339	0.796	0.476
Baseline + spaCy Filter	0.345	0.796	0.481
Hybrid (Baseline + Llama 3)	0.602	0.814	0.692

The introduction of Llama 3 as a semantic judge drastically improves Precision. By instructing the LLM to specifically split chimeras formed by the lack of ASR punctuation, the pipeline recovers distinct entities that would have otherwise been misclassified.

Furthermore, the fuzzy matching evaluation reveals the true efficacy of the pipeline: the system successfully identified over 81% of the named entities (Recall) despite the phonetic noise and overlaps of the television broadcast. By tolerating minor ASR spelling variations, the F1-Score accurately reflects the tool’s high utility for downstream archiving and indexing tasks, where an 85% spelling accuracy on a proper noun is largely sufficient for searchability. Finally, pre-filtering the candidates via the CamemBERT/spaCy baseline reduced the prompt size sent to the LLM, cutting token consumption and latency drastically compared to raw transcript processing.

4.5 Error Analysis

While the hybrid pipeline significantly outperforms the baseline, a qualitative analysis of the remaining False Positives (FP) and False Negatives (FN) reveals three persistent challenges inherent to spoken language Information Extraction:

- **Metonymy and Semantic Ambiguity (LOC vs. ORG):** In broadcast news, the boundary between a geographical location (LOC) and a political entity (ORG) is highly contextual. For instance, a state like "Israël" might be used to describe a territory in one sentence and an acting government in the next. Slight discrepancies between the human annotator's interpretation and the LLM's classification account for a notable share of label mismatches.
- **Phonetic Fragmentation of Proper Nouns:** ASR systems frequently struggle with non-native or complex names, causing the downstream NER model to shatter a single entity into multiple fragmented tokens. For example, the ground truth entity "Marina Bekkaoui" was transcribed and subsequently split into two distinct PER entities ("Marina" and "Mekawi"). Under our evaluation framework, this phenomenon triggers a triple penalty: generating two FPs for the fragments and one FN for the missed full name, thereby artificially deflating the Precision score.
- **Over-Aggressive LLM Filtering:** Although the LLM excels at removing conversational noise and false positives, it occasionally acts too aggressively when evaluating metonymies. Institutional entities that double as physical buildings or common nouns, such as the "Maison Blanche" or the "Bundestag," are sometimes downgraded by the model's strict semantic judgment and dropped from the final output, leading to unwarranted False Negatives.

5 Conclusion

In this paper, we addressed the critical bottleneck of performing robust Named Entity Recognition on noisy, unpunctuated television broadcast audio. While standard NLP models fail catastrophically when confronted with ASR-induced entity fusion, and Large Language Models remain too computationally expensive for raw transcript processing at scale, our proposed hybrid methodology bridges this gap effectively.

By disabling default aggregation strategies and implementing a strict token-level BIO rule set, we successfully decapitated semantic chimeras early in the pipeline. Furthermore, our context-aware grammatical filter eliminated the vast majority of systemic false positives. This heavy baseline filtering allowed us to utilize Llama 3 strictly as a semantic judge, deploying it only on highly probable candidates. As demonstrated in our experiments on the Golden Dataset, this approach achieved a massive reduction in API token consumption while successfully splitting complex overlapping entities and dramatically raising the overall Precision of the pipeline. Finally, our fault-tolerant batch-processing architecture ensures that this solution is highly scalable and production-ready for massive audiovisual archives.

5.1 Limitations and Future Work

Despite the promising results, our approach has several limitations that warrant future investigation. First, the dependency on an external API (Groq) for the LLM refinement stage introduces latency and rate-limit bottlenecks. A natural progression would involve fine-tuning a much smaller, quantized generative model (e.g., a 2B or 3B parameter architecture) specifically on our Golden Dataset. This model could be run entirely locally alongside CamemBERT, completely eliminating network dependencies and HTTP 429 constraints.

Secondly, while the current architecture evaluates entities exclusively at the document level, future work will focus on strict temporal alignment. By preserving the exact timestamp outputs from Faster-Whisper and mapping them through the CamemBERT tokenization process, the system could produce highly granular JSON records indicating the precise millisecond an entity is spoken. This would unlock advanced downstream applications, such as automated video indexing and smart-chapter generation for multimedia platforms.

Fig. 1 Overview of the proposed hybrid extraction pipeline: From raw audio processing to LLM semantic refinement and checkpoint evaluation.

References

- [1] Li, J., Sun, A., Han, J., Li, C.: A survey on deep learning for named entity recognition. *IEEE Transactions on Knowledge and Data Engineering* **34**(1), 50–70 (2020)
- [2] Martin, L., Muller, B., Suárez, P.J.O., Dupont, Y., Romary, L., Clergerie, É.V., Seddah, D., Sagot, B.: Camembert: a tasty french language model. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 7203–7219 (2020)
- [3] Yamada, I., Asai, A., Shindo, H., Takeda, H., Matsumoto, Y.: Luke: Deep contextualized entity representations with entity-aware self-attention. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6442–6454 (2020)
- [4] Jiang, Z., Xu, W., Araki, J., Neubig, G.: Generalizing natural language analysis through span-relation representations. *arXiv preprint arXiv:2004.13715* (2020)
- [5] Wang, X., Jiang, Y., Bach, N., Wang, T., Huang, Z., Huang, F., Tu, K.: Automated concatenation of embeddings for structured prediction. In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, pp. 2643–2660 (2021)
- [6] Le, H., Vial, L., Frej, J., Segonne, V., Coavoux, M., Lecouteux, B., Allauzen, A., Crabbé, B., Besacier, L., Schwab, D.: Flaubert: Unsupervised language model

- pre-training for french. In: Proceedings of the 12th Language Resources and Evaluation Conference, pp. 2469–2474 (2020)
- [7] Suárez, P.J.O., Romary, L., Sagot, B.: A monolingual approach to contextualized word embeddings for mid-resource languages. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 1703–1714 (2020)
 - [8] Fu, J., Liu, P., Neubig, G.: Rethinking generalization of neural models: A named entity recognition case study. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, pp. 7732–7739 (2020)
 - [9] Bernier-Colborne, G., Langlais, P.: Hardeval: Focusing on challenging tokens to assess robustness of ner. In: Proceedings of the 12th Language Resources and Evaluation Conference, pp. 1–9 (2020)
 - [10] Ushio, A., Camacho-Collados, J.: T-ner: An all-round python library for transformer-based named entity recognition. In: Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics, pp. 53–62 (2021)
 - [11] Weng, Y., Mirsamadi, S., Ganapathiraju, A.: Joint contextual modeling for asr correction and spoken language understanding. In: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 7999–8003 (2020). IEEE
 - [12] Lin, B., Xu, W.: Spoken language understanding: A survey. arXiv preprint arXiv:2104.01511 (2021)
 - [13] Ding, N., Xu, G., Chen, Y., Wang, X., Han, X., Xie, P., Zheng, H., Liu, Z.: Fewnerd: A few-shot named entity recognition dataset. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 3198–3213 (2021)
 - [14] Liu, Z., Yan, Y., Mukherjee, A., Ter-Sarkisov, A.: Crossner: Evaluating cross-domain named entity recognition. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, pp. 13452–13460 (2021)
 - [15] Desot, T., Portet, F., Vacher, M.: Named entity recognition for spoken french. In: Proceedings of the 12th Language Resources and Evaluation Conference, pp. 3456–3464 (2020)
 - [16] Ghannay, S., Caubrière, A., Estève, Y., Camelin, N., *et al.*: Impact of asr errors on information extraction. IEEE/ACM Transactions on Audio, Speech, and Language Processing **28**, 1123–1135 (2020)
 - [17] Caubrière, A., Tomashenko, N., Laurent, A., Morin, E., Camelin, N., Estève, Y.: End-to-end named entity extraction from speech. arXiv preprint

arXiv:1805.12045 (2018)

- [18] Radford, A., Kim, J.W., Xu, T., Brockman, G., McLeavey, C., Sutskever, I.: Robust speech recognition via large-scale weak supervision. In: International Conference on Machine Learning, pp. 28492–28518 (2023). PMLR
- [19] Baevski, A., Zhou, Y., Mohamed, A., Auli, M.: wav2vec 2.0: A framework for self-supervised learning of speech representations. In: Advances in Neural Information Processing Systems, vol. 33, pp. 12449–12460 (2020)
- [20] Hsu, W.-N., Bolte, B., Tsai, Y.-H.H., Lakhota, K., Salakhutdinov, R., Mohamed, A.: Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **29**, 3451–3460 (2021)
- [21] Chen, S., Wang, C., Chen, Z., Wu, Y., Liu, S., Chen, Z., Li, J., Kanda, N., Yoshioka, T., Xiao, X., *et al.*: Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing* **16**(6), 1505–1518 (2022)
- [22] Qin, L., Che, W., Li, Y., Wen, H., Ni, J.: A survey on spoken language understanding: Recent advances and new frontiers. *arXiv preprint arXiv:2103.03095* (2021)
- [23] Arora, S., *et al.*: Jointly learning to align and translate with transformer models for asr. In: Proceedings of Speech Communication (2022)
- [24] Courtland, M., Faulkner, A., McElvain, G.: Punctuation restoration for audio transcripts. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 1–10 (2020)
- [25] Alam, T., *et al.*: Robust punctuation restoration in spoken language. *Journal of Audio Engineering* **45** (2020)
- [26] Park, S., Albin, A., Ubale, R.: A punctuation restoration system for l2 speech using text and acoustic features. In: ISCA Symposium on Speech and Language Technology in Education (SLaTE), pp. 56–60 (2023)
- [27] Saxon, M., Choudhary, S., McKenna, J.P., Mouchtaris, A.: End-to-end spoken language understanding for generalized voice assistants. In: Interspeech, pp. 4738–4742 (2021)
- [28] Sun, G., Zhang, C., Woodland, P.F.: End-to-end spoken language understanding with tree-constrained pointer generator. In: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1–5 (2023)

- [29] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., *et al.*: Language models are few-shot learners. In: Advances in Neural Information Processing Systems, vol. 33, pp. 1877–1901 (2020)
- [30] Wang, S., Sun, X., Li, X., Ouyang, R., Wu, F., Tian, H., Liu, W.: Gpt-ner: Named entity recognition via large language models. arXiv preprint arXiv:2304.10428 (2023)
- [31] OpenAI: Gpt-4 technical report. arXiv preprint arXiv:2303.08774 (2023)
- [32] AI, M.: Llama 3 technical report. arXiv preprint arXiv:2404.XXXXX (2024)
- [33] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q.V., Zhou, D.: Chain-of-thought prompting elicits reasoning in large language models. In: Advances in Neural Information Processing Systems, vol. 35, pp. 24824–24837 (2022)